Name: ..... Due: **Next Class / Friday**

C3 The Electric Guitar HW 3 – Eddie Van Halen 'Eruption', Floyd Rose & Digital Amp Modelling

Artist: Van Halen (Van Halen, 1978, Guitarist: Eddie Van Halen, Producer: Ted Templeman)
https://www.youtube.com/watch?v=M4Czx8EWXb0

1. "Eruption" redefined modern rock guitar technique and timbre. Identify the custom hardware elements below. (3)

Technical Element	Custom Modification & Sonic Function
Frankenstrat DIY Hybrid	Stratocaster body retrofitted with a slanted Gibson PAF humbucker direct to si
Variac Voltage Starvation	Variable transformer reducing AC mains to ~90V; lowers tube plate voltage fo
MXR Phase 90 Pedal	
Univox EP-3 Echoplex Tape Echo	

2. Explain how a Floyd Rose double-locking tremolo system prevents strings from slipping out of tune during extreme dive-bombs and pitch pull-ups. (3)

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3. What are active pickups (e.g. EMG 81), and what technical advantages do their low-impedance preamp circuitry offer over traditional passive pickups? (3)

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4. Explain the difference between digital tube amplifier simulation algorithms (DSP waveshaping/component modeling) and Speaker Cabinet Impulse Responses (IR convolution). (3)

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5. Direct Injection (DI) & Amp Profiling: Explain how modern profilers (Kemper, Quad Cortex) allow guitarists to record silent studio-quality tracks without physical speaker cabinets or microphone bleed. (3)

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6. Traditional Tube Half-Stack vs Modern Digital Modeller / Profiler: compare below: (4)

100W Tube Amplifier & 4x12 Cabinet (Microphoned in Studio) (2)
Valve saturation, acoustic sound pressure, room reflections, microphone placement proximity effect:
Digital DSP Modeller & Convolution IR (Direct to DAW) (2)
Mathematical modeling, zero bleed, total recall, silent recording, and identical repeatable tone:

Total:	**/18**